

Inner Product Spaces & Gram-Schmidt Orthogonalization Process

Linear Algebra

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Motivation

- Geometry of vector spaces
- Measuring lengths and angles
- Constructing “nice” bases

Inner Product

Definition (Inner Product)

An inner product on a vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\alpha \in \mathbb{R}$:

- **Linearity:** $\langle \alpha \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle = \underbrace{\langle \alpha \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle}$
- **Symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
- **Positivity:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Remark

The inner product, also known as the dot product, provides us with geometric concepts such as the length of a vector, the angle between two vectors, and orthogonality. We will extend these concepts to abstract vector spaces so that geometric ideas can be used to describe abstract vectors.

$$\langle u+v, w+z \rangle = \langle w+z, u+v \rangle$$

$$u, v, w, z \in V \rightarrow \langle u+v, w+z \rangle = \langle u, w+z \rangle + \langle v, w+z \rangle$$

$$= \langle w+z, u \rangle + \langle w+z, v \rangle$$

$$= \langle w, u \rangle + \langle z, u \rangle + \langle w, v \rangle + \langle z, v \rangle$$

$$\langle u+v, \underbrace{w+z}_w \rangle = \langle u, w \rangle + \langle u, z \rangle + \langle v, w \rangle + \langle v, z \rangle$$

$$= \langle w+z, u+v \rangle$$

Inner Product Spaces

Definition (Inner Product Spaces)

The vector space V with an inner product is called a (real) inner product space.

Example

For $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \in \mathbb{R}^2$, we define

$$\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 - x_1y_2 - x_2y_1 + 5x_2y_2.$$

Then $\langle \cdot, \cdot \rangle$ is an inner product.

$\forall x, y, z \in \mathbb{R}^2$

$$\langle \alpha \mathbf{x} + \mathbf{z}, \mathbf{y} \rangle = \left\langle \alpha \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{z}, \mathbf{y} \rangle$$

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} \alpha x_1 + z_1 \\ \alpha x_2 + z_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \Rightarrow \langle \begin{pmatrix} \alpha x_1 + z_1 \\ \alpha x_2 + z_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \rangle = 2(\alpha x_1 + z_1)y_1 - (\alpha x_1 + z_1)y_2 - (\alpha x_2 + z_2)y_1 + 5(\alpha x_2 + z_2)y_2$$

(check)

Ex: $\mathbb{R}^2$, $\langle x, y \rangle = x_1 y_1 + x_2 y_2$



DPS

$$\begin{aligned}
 \langle x, x \rangle &= \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right\rangle = 2x_1^2 - x_1x_2 - x_2x_1 + 5x_2^2 \\
 &= \left(\sqrt{2}x_1 - \frac{1}{\sqrt{2}}x_2 \right)^2 + 5x_2^2 - \frac{1}{2}x_2^2 \\
 &= \left(\sqrt{2}x_1 - \frac{1}{\sqrt{2}}x_2 \right)^2 + \left(\frac{9}{2} \right)x_2^2
 \end{aligned}$$

$$\langle x, x \rangle = 0 \Leftrightarrow \begin{matrix} \geq 0 \\ \sqrt{2}x_1 - \frac{1}{\sqrt{2}}x_2 = 0 \text{ and } x_2 = 0 \end{matrix}$$

$$\Leftrightarrow \sqrt{2}x_1 = \frac{1}{\sqrt{2}}x_2 = 0. \odot$$

$$\Rightarrow x_1 = 0 \text{ and } x_2 = 0$$

Examples of IPS

Example

The vector space $C[a, b]$ of all real valued continuous functions on a closed interval $[a, b]$ is an inner product space with the inner product:

$$\langle f, g \rangle = \int_a^b f(t)g(t)dt, \quad f, g \in C[a, b].$$

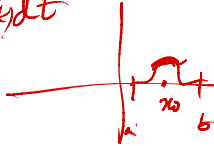
Handwritten notes:

$$\langle f, f \rangle = \int_a^b f^2(t)dt$$

IPS $\Rightarrow \int_a^b f^2(t)dt = 0$

Suppose f is not zero throughout. $f(x_0) \neq 0$.

$f^2(x_0) \neq 0 \Rightarrow \boxed{f^2(x_0) > 0}$



Examples of IPS

Example

The vector space $\mathcal{M}_{m,n}(\mathbb{R})$ of all real matrices with size $m \times n$ is an inner product space with the inner product:

$$\langle A, B \rangle = \text{tr}(B^T A) \text{ for all } A, B \in \mathcal{M}_{m,n}(\mathbb{R}).$$

* sym: $\langle B, A \rangle = \text{tr}(A^T B) = \text{tr}(B^T A) = \langle A, B \rangle = \text{tr}(A^T B^T)$

$$\langle A, A \rangle = \text{tr}(A^T A) \geq 0?$$

IPS

Examples of IPS

Example

The vector space $\mathbb{R}^n$ with the inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

is an inner product space.

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Representation of IP

- Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis of an n -dimensional inner product space V . For vectors $\mathbf{u}, \mathbf{v} \in V$, write

$$\mathbf{u} = x_1 \mathbf{u}_1 + x_2 \mathbf{u}_2 + \cdots + x_n \mathbf{u}_n,$$

$$\mathbf{v} = y_1 \mathbf{u}_1 + y_2 \mathbf{u}_2 + \cdots + y_n \mathbf{u}_n.$$

- By linearity of the inner product,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \left\langle \sum_{i=1}^n x_i \mathbf{u}_i, \sum_{j=1}^n y_j \mathbf{u}_j \right\rangle = \sum_{i=1}^n \sum_{j=1}^n x_i y_j \langle \mathbf{u}_i, \mathbf{u}_j \rangle.$$

$$\begin{aligned} &= \langle x_1 \mathbf{u}_1 + x_2 \mathbf{u}_2 + \cdots + x_n \mathbf{u}_n, \sum y_j \mathbf{u}_j \rangle \\ &= \langle x_1 \mathbf{u}_1, \sum y_j \mathbf{u}_j \rangle + \langle x_2 \mathbf{u}_2, \sum y_j \mathbf{u}_j \rangle + \cdots + \langle x_n \mathbf{u}_n, \sum y_j \mathbf{u}_j \rangle \\ &= x_1 \langle \mathbf{u}_1, \sum y_j \mathbf{u}_j \rangle + x_2 \langle \mathbf{u}_2, \sum y_j \mathbf{u}_j \rangle + \cdots + x_n \langle \mathbf{u}_n, \sum y_j \mathbf{u}_j \rangle \end{aligned}$$

Representation of IP: Contd.

- We call the $n \times n$ matrix

$$A = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_1, \mathbf{u}_n \rangle \\ \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_2, \mathbf{u}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_n, \mathbf{u}_1 \rangle & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}$$

the matrix of the inner product $\langle \cdot, \cdot \rangle$ relative to the basis $\mathcal{B}$.

- Thus, using coordinate vectors

$$[\mathbf{u}]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad [\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix},$$

we have

$$\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{u}]_{\mathcal{B}}^T A [\mathbf{v}]_{\mathcal{B}}.$$

Representation of IP

Theorem

Let V be an n -dimensional vector space with an inner product $\langle \cdot, \cdot \rangle$, and let A be the matrix of $\langle \cdot, \cdot \rangle$ relative to a basis $\mathcal{B}$. Then, for any vectors $\mathbf{u}, \mathbf{v} \in V$

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{x}^T A \mathbf{y}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the coordinate vectors of $\mathbf{u}, \mathbf{v}$, respectively, i.e., $\mathbf{x} = [\mathbf{u}]_{\mathcal{B}}$ and $\mathbf{y} = [\mathbf{v}]_{\mathcal{B}}$.

Properties of the Matrix of the IP

Geometric Interpretation

- 1 The matrix A encodes the geometry of the space in the chosen basis.
- 2 If the basis is not orthonormal, A corrects for non-orthogonality.
- 3 Changing basis changes A , but the inner product itself remains the same.

Change of Bases

- Let V be an n -dimensional inner product space. Suppose

$$\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}, \quad \mathcal{B}' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$$

are two bases of V .

- Let

$$A = (\langle \mathbf{u}_i, \mathbf{u}_j \rangle)$$

be the matrix of the inner product relative to the basis $\mathcal{B}$, and

$$A' = (\langle \mathbf{v}_i, \mathbf{v}_j \rangle)$$

the matrix relative to $\mathcal{B}'$.

- Let P be the change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$, defined by

$$[\mathbf{v}]_{\mathcal{B}} = P[\mathbf{v}]_{\mathcal{B}'}$$

- Equivalently, the columns of P are given by

$$P = \begin{bmatrix} [\mathbf{v}_1]_{\mathcal{B}} & [\mathbf{v}_2]_{\mathcal{B}} & \cdots & [\mathbf{v}_n]_{\mathcal{B}} \end{bmatrix}.$$

Change of Bases: Contd.

- We know that for vectors $\mathbf{x}, \mathbf{y} \in V$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_{\mathcal{B}}^T A [\mathbf{y}]_{\mathcal{B}}.$$

- Using the change of basis,

$$[\mathbf{x}]_{\mathcal{B}} = P[\mathbf{x}]_{\mathcal{B}'}, \quad [\mathbf{y}]_{\mathcal{B}} = P[\mathbf{y}]_{\mathcal{B}'}$$

- Substituting,

$$\langle \mathbf{x}, \mathbf{y} \rangle = (P[\mathbf{x}]_{\mathcal{B}'})^T A (P[\mathbf{y}]_{\mathcal{B}'}).$$

- Using the transpose property,

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_{\mathcal{B}'}^T P^T A P [\mathbf{y}]_{\mathcal{B}'}.$$

- But also,

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_{\mathcal{B}'}^T A' [\mathbf{y}]_{\mathcal{B}'}.$$

- Therefore,

$$\boxed{A' = P^T A P}$$

where P^* denotes the conjugate transpose of P .

Norm of a Vector

IP. $\langle v, v \rangle > 0$ if $v \neq 0$.

Definition (Norm of a Vector)

For each vector $\mathbf{u} \in V$, the norm (also called the length) of $\mathbf{u}$ is defined as the number $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$.

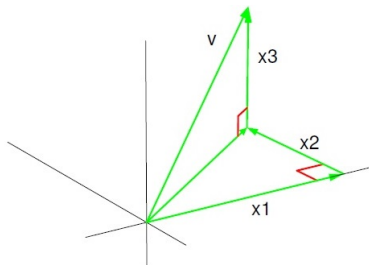


Figure: The length of a vector¹ $\mathbb{R}^3$

¹Source: <https://url-shortener.me/C2Z2>

Properties of Norm

Theorem

The norm in an inner product space V satisfies the following properties:

(N1) $\|\mathbf{v}\| \geq 0$; and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

(N2) $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$, for all $\mathbf{v} \in V$.

(N3) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, $\mathbf{u}, \mathbf{v} \in V$.

↳ triangular ineq.

Cauchy-Schwarz Inequality

Definition (Cauchy-Schwarz Inequality)

For any vectors $\mathbf{u}, \mathbf{v}$ in an inner product space V ,

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle.$$

Equivalently,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

$\Downarrow$

$\Rightarrow -\|\mathbf{u}\| \|\mathbf{v}\| \leq \langle \mathbf{u}, \mathbf{v} \rangle \leq \|\mathbf{u}\| \|\mathbf{v}\|$

The Concept of Angle B/W Two Non-Zero Vectors

For nonzero vectors $\mathbf{u}, \mathbf{v} \in V$, the Cauchy-Schwarz inequality implies

$-1 \leq \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1$. The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is defined by

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

$\|\mathbf{u}\| \neq 0$
 $\|\mathbf{v}\| \neq 0$

The angle exists and is unique.

Orthogonality

Definition (Orthogonal/Orthonormal Vectors)

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are called orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ and is denoted as $\mathbf{u} \perp \mathbf{v}$. We call a set of vectors *Orthogonal Set*, if all distinct pairs are orthogonal. Moreover, we call a vector *Orthonormal* if it is orthogonal and of unit length. Also, a set of vectors *Orthonormal Set* if they are orthogonal and of unit length.

Theorem (Pythagorean Theorem)

Let V be an inner product space. If $\mathbf{u}, \mathbf{v} \in V$ and $\mathbf{u} \perp \mathbf{v}$. Then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

$$\begin{aligned} \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle &= \|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle + \cancel{\langle \mathbf{u}, \mathbf{v} \rangle} + \cancel{\langle \mathbf{v}, \mathbf{u} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \end{aligned}$$

Triangle Inequality

Definition (Triangle Inequality)

Let V be an inner product space. Then, for all $\mathbf{u}, \mathbf{v} \in V$ we have

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

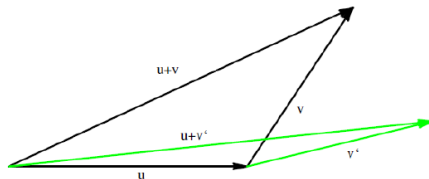


Figure: Triangle Inequality ²

²Source: <https://url-shortener.me/C2Z2>

Parallelogram Law/Identity

Theorem (Parallelogram Law)

Let V be an inner product space. Then, for all $\mathbf{u}, \mathbf{v} \in V$ we have

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2).$$

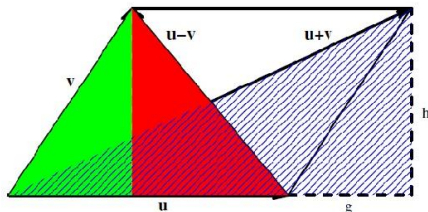


Figure: ~~Triangle Inequality~~³

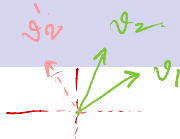
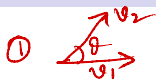
³Source: <https://url-shortener.me/C2Z2>

$$\textcircled{A} \quad \|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2.$$

$$\textcircled{B} \quad \|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\ = \|u\|^2 - 2\langle u, v \rangle + \|v\|^2.$$

$$\textcircled{A} + \textcircled{B} \Rightarrow \boxed{\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)}$$

Orthonormal Basis



$\{v_1, v_2\} \rightarrow \nabla$
 $\{v_1, v_2'\} \rightarrow \perp$

Definition (Orthonormal Basis)

An orthonormal basis of a finite-dimensional inner product space is a list of orthonormal vectors that is basis for V .

Remark

Let $\{e_1, \dots, e_n\}$ is **orthonormal** then

$$\boxed{v = \sum_{i=1}^n \langle v, e_i \rangle e_i} \quad \text{di.} \quad \text{or} \quad \sum_{i=1}^n \boxed{\alpha_i} v_i$$

and

$$\|v\|^2 = \sum_{i=1}^n \|\langle v, e_i \rangle\|^2.$$

Suppose $V \rightarrow$ finite dimensional $\dim(V) = n$.

$$\boxed{\alpha_1 = \langle v, e_1 \rangle}$$

$$\alpha_2 = \langle v, e_2 \rangle$$

$$\alpha_i = \langle v, e_i \rangle$$

$\{e_1, e_2, \dots, e_n\}$ is a basis which is orthonormal

$$\langle e_i, e_j \rangle = \delta_{ij} \longrightarrow \text{Kronecker's delta}$$

$$= \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

Paul Dirac

Let $v \in V$.

$$\Rightarrow v = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, \quad (\alpha_i \in \mathbb{R})$$

$i=1(1)n$

$$\begin{aligned} \langle v, e_i \rangle &= \langle \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, e_i \rangle \\ &= \langle \alpha_1 e_1, e_i \rangle + \langle \alpha_2 e_2, e_i \rangle + \dots + \langle \alpha_n e_n, e_i \rangle \\ &= \alpha_1 \langle e_1, e_i \rangle + \alpha_2 \langle e_2, e_i \rangle + \dots + \alpha_n \langle e_n, e_i \rangle \end{aligned}$$

$$\|v\|^2 = \langle v, v \rangle$$

$$= \langle \underbrace{\langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n}_{n\text{-terms}}, \underbrace{\langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n}_{n\text{-terms}} \rangle$$

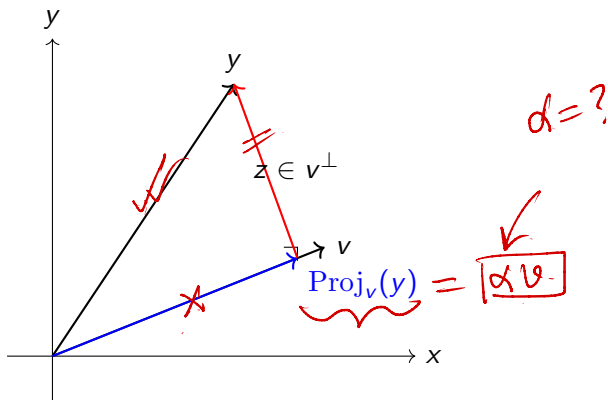
$$= \langle \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n}_{n\text{-terms}}, \underbrace{\langle v, e_1 \rangle e_1}_{*} \rangle + \langle \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n}_{n\text{-terms}}, \underbrace{\langle v, e_2 \rangle e_2}_{*} \rangle + \dots + \langle \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n}_{n\text{-terms}}, \underbrace{\langle v, e_n \rangle e_n}_{*} \rangle$$

$$= \langle \langle v, e_1 \rangle e_1, \langle v, e_1 \rangle e_1 \rangle + \langle \langle v, e_2 \rangle e_2, \langle v, e_1 \rangle e_1 \rangle + \dots + \langle \langle v, e_n \rangle e_n, \langle v, e_1 \rangle e_1 \rangle +$$

$$= \langle v, e_1 \rangle \langle v, e_1 \rangle \langle e_1, e_1 \rangle + \langle v, e_2 \rangle \langle v, e_1 \rangle \langle e_2, e_1 \rangle + \dots + \langle v, e_n \rangle \langle v, e_1 \rangle \langle e_n, e_1 \rangle +$$

$$= \langle v, e_1 \rangle^2 + 0 + \dots + 0 + \langle v, e_2 \rangle^2 + 0 + \dots + 0 + \langle v, e_n \rangle^2$$

Geometric Interpretation of Orthogonal Projection



$$y = \text{Proj}_v(y) + z, \quad \boxed{z \perp v}$$
$$= \alpha v + z = \frac{\langle v, y \rangle}{\|v\|^2} v + z = y$$

Orthogonal Decomposition

- Let V be an inner product space and let $\mathbf{v} \in V$ be a nonzero vector. We want to decompose an arbitrary vector $\mathbf{y} \in V$ as

$$\mathbf{y} = \alpha \mathbf{v} + \mathbf{z}, \quad \text{where } \mathbf{z} \perp \mathbf{v}.$$

- Since $\mathbf{z} \perp \mathbf{v}$, we have

$$\langle \mathbf{v}, \mathbf{y} \rangle = \langle \mathbf{v}, \alpha \mathbf{v} + \mathbf{z} \rangle = \alpha \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{z} \rangle = \alpha \langle \mathbf{v}, \mathbf{v} \rangle.$$

- Thus,

$$\alpha = \frac{\langle \mathbf{v}, \mathbf{y} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} = \frac{\langle \mathbf{v}, \mathbf{y} \rangle}{\|\mathbf{v}\|^2}.$$

$$\text{Proj}_{\mathbf{v}}(\mathbf{y}) = \frac{\langle \mathbf{v}, \mathbf{y} \rangle}{\|\mathbf{v}\|^2} \mathbf{v}.$$

Orthogonal Projection onto a Vector

Definition (Orthogonal Projection onto a Vector)

The orthogonal projection of $\mathbf{y}$ onto the direction of $\mathbf{v}$ is

$$\text{Proj}_{\mathbf{v}}(\mathbf{y}) = \frac{\langle \mathbf{v}, \mathbf{y} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}.$$

The linear transformation $\text{Proj}_{\mathbf{v}} : V \rightarrow V$ is called the orthogonal projection onto the direction $\mathbf{v}$.

Projection in $\mathbb{R}^n$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\langle x, y \rangle = x^T y.$$

$$\langle v, v \rangle = v^T v = \|v\|^2$$

Proposition

Let $\mathbf{v}$ be a nonzero vector in $\mathbb{R}^n$. Then, there holds

$$\text{Proj}_{\mathbf{v}}(\mathbf{y}) = \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} (\mathbf{v} \mathbf{v}^T) \mathbf{y}.$$

The orthogonal projection onto $\mathbf{v}^\perp$ is

$$\text{Proj}_{\mathbf{v}^\perp}(\mathbf{y}) = \left(I - \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T \right) \mathbf{y} = (I - \text{Proj}_{\mathbf{v}}) \mathbf{y}$$

Matrix Form of the Projection

- Write

$$v = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}.$$

- Then

$$vv^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

- v is an $n \times 1$ matrix.
- v^T is a $1 \times n$ matrix.
- vv^T is an $n \times n$ matrix.

$v^T v$ is a scalar (1×1)

Matrix Form of the Projection: Contd.

- Now we know that

$$\begin{aligned}\text{Proj}_{\mathbf{v}}(\mathbf{y}) &= \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \langle \mathbf{v}, \mathbf{y} \rangle \mathbf{v} \\ &= \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \langle \mathbf{v}, \mathbf{y} \rangle \\ &= \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T \mathbf{y}.\end{aligned}$$

- Hence the standard matrix of $\text{Proj}_{\mathbf{v}}$ is

$$\boxed{\text{Proj}_{\mathbf{v}} = \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T.}$$

Projection onto the Orthogonal Complement

- Since

$$\text{Proj}_{\mathbf{v}^\perp}(\mathbf{y}) = \mathbf{y} - \text{Proj}_{\mathbf{v}}(\mathbf{y}),$$

we obtain

$$\text{Proj}_{\mathbf{v}^\perp}(\mathbf{y}) = \left(I - \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T \right) \mathbf{y}.$$

- Thus the standard matrix of $\text{Proj}_{\mathbf{v}^\perp}$ is

$$I - \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T.$$

Example

Example

Find the linear mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$ that is the orthogonal projection of $\mathbb{R}^3$ onto the plane

$$x_1 + x_2 + x_3 = 0.$$

Example

Example

Find the linear mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$ that is the orthogonal projection of $\mathbb{R}^3$ onto the plane

$$x_1 + x_2 + x_3 = 0.$$

→ n.v. (1,1,1)

Solution: To find the orthogonal projection of $\mathbb{R}^3$ onto the subspace $\mathbf{v}^\perp$, where $\mathbf{v} = [1, 1, 1]^T$. Then, the orthogonal projection of $\mathbf{y}$ onto $\mathbf{v}^\perp$ is given by

$$\begin{aligned}\text{Proj}_{\mathbf{v}^\perp}(\mathbf{y}) &= \left(I_3 - \frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} \mathbf{v}^T \right) \mathbf{y} \\ &= \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (1 \ 1 \ 1) \right\} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.\end{aligned}$$

vector

3x1

Orthogonal Decomposition

- Let V be an inner product space and W a subspace of V . Suppose $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an **orthogonal basis** of W .
- For any $\mathbf{y} \in V$, we want to decompose

$$\mathbf{y} = \mathbf{w} + \mathbf{z}$$

where

$$\mathbf{w} \in W, \quad \mathbf{z} \in W^\perp.$$

- Since $w \in W$, there exist scalars $\alpha_1, \dots, \alpha_k$ such that

$$\mathbf{w} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k.$$

Finding the Coefficients

- Because $\mathbf{z} \perp \mathbf{v}_i$ for all i , we have

$$\langle \mathbf{v}_i, \mathbf{y} \rangle = \langle \mathbf{v}_i, \mathbf{w} + \mathbf{z} \rangle = \langle \mathbf{v}_i, \mathbf{w} \rangle.$$

- Using orthogonality:

$$\langle \mathbf{v}_i, \mathbf{w} \rangle = \alpha_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle.$$

- Hence,

$$\alpha_i = \frac{\langle \mathbf{v}_i, \mathbf{y} \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle}, \quad 1 \leq i \leq k.$$

Orthogonal Projection Formula

- We define the **orthogonal projection** of y onto W by

$$\text{Proj}_W(\mathbf{y}) = \sum_{i=1}^k \frac{\langle \mathbf{v}_i, \mathbf{y} \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i.$$

- The component orthogonal to W is

$$\text{Proj}_{W^\perp}(\mathbf{y}) = \mathbf{y} - \text{Proj}_W(\mathbf{y}).$$

Special Case: Orthonormal Basis

- If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an **orthonormal basis** of W , then

$$\langle \mathbf{v}_i, \mathbf{v}_i \rangle = 1.$$

- So the projection simplifies to:

$$\text{Proj}_W(\mathbf{y}) = \sum_{i=1}^k \langle \mathbf{v}_i, \mathbf{y} \rangle \mathbf{v}_i.$$

Orthogonal Projection of V onto W

Theorem (Orthogonal Projection of V onto W)

Let V be an n -dimensional inner product space and W a subspace with orthogonal basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Then for any $\mathbf{y} \in V$,

$$\text{Proj}_W(\mathbf{y}) = \sum_{i=1}^k \frac{\langle \mathbf{v}_i, \mathbf{y} \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i,$$

$$\text{Proj}_{W^\perp}(\mathbf{y}) = \mathbf{y} - \text{Proj}_W(\mathbf{y}). \Rightarrow \mathbf{y} = \text{Proj}_W(\mathbf{y}) + \text{Proj}_{W^\perp}(\mathbf{y})$$

In particular, if $\mathcal{B}$ is an orthonormal basis of W , then

$$\text{Proj}_W(\mathbf{y}) = \sum_{i=1}^k \langle \mathbf{v}_i, \mathbf{y} \rangle \mathbf{v}_i.$$

Matrix Form of Projection

Proposition: Projection Matrix

Let W be a subspace of $\mathbb{R}^n$. Let

$$\dim(W) = k$$

$$U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_k]$$

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$$

be an $n \times k$ matrix whose columns form an **orthonormal basis** of W .
Then the orthogonal projection is given by

$$\text{Proj}_W(\mathbf{y}) = UU^T \mathbf{y}.$$

Remark

The matrix $P = UU^T$ is called the **projection matrix** onto W .

Properties of the Projection Matrix

Properties of the Projection Matrix

If $P = UU^T$, then:

- $P^2 = P$ (idempotent)
- $P^T = P$ (symmetric)
- $\text{Range}(P) = W$
- $\text{Null}(P) = W^\perp$

} Proof

Setup: Gram-Schmidt Orthogonalization Process

- Let W be a subspace of an inner product space V . Let

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

be a basis of W (not necessarily orthogonal).

- Goal:* Construct an **orthogonal basis**

$$B' = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$$

$$w_i \perp w_j$$

for W .

Step-by-Step Construction

- Define recursively:

$$\mathbf{w}_1 = \mathbf{v}_1, \quad W_1 = \text{Span}\{\mathbf{w}_1\}$$

$$\mathbf{w}_2 = \mathbf{v}_2 - \text{Proj}_{W_1}(\mathbf{v}_2), \quad W_2 = \text{Span}\{\mathbf{w}_1, \mathbf{w}_2\}$$

$$\mathbf{w}_3 = \mathbf{v}_3 - \text{Proj}_{W_2}(\mathbf{v}_3)$$

$$\vdots$$

$$\mathbf{w}_k = \mathbf{v}_k - \text{Proj}_{W_{k-1}}(\mathbf{v}_k)$$

Explicit Formula

- More precisely,

$$\mathbf{w}_1 = \mathbf{v}_1$$

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{w}_1, \mathbf{v}_2 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1$$

$$\mathbf{w}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{w}_1, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 - \frac{\langle \mathbf{w}_2, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2$$

$\vdots$

$$\mathbf{w}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{w}_j, \mathbf{v}_k \rangle}{\langle \mathbf{w}_j, \mathbf{w}_j \rangle} \mathbf{w}_j$$

$$\mathbf{w}_1 \perp \mathbf{w}_2$$

$$\langle \mathbf{w}_1, \mathbf{w}_2 \rangle = 0$$

$$\langle \mathbf{v}_1, \mathbf{v}_2 - \frac{\langle \mathbf{w}_1, \mathbf{v}_2 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 \rangle$$

$= 0$

inner
product
defⁿ

Imp

Span Preservation

- Each $\mathbf{w}_i$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_i$.
- Conversely,

$$\mathbf{v}_1 = \mathbf{w}_1$$

$$\mathbf{v}_2 = \frac{\langle \mathbf{w}_1, \mathbf{v}_2 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 + \mathbf{w}_2$$

$$\mathbf{v}_3 = \frac{\langle \mathbf{w}_1, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 + \frac{\langle \mathbf{w}_2, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2 + \mathbf{w}_3$$

$\vdots$

- Hence,

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_k\}.$$



Conclusion

- Since $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a basis of W ,

$$\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$$

is also a basis of W .

- Moreover, the vectors $\mathbf{w}_1, \dots, \mathbf{w}_k$ are **orthogonal**.
- This procedure is called the

Gram–Schmidt Orthogonalization Process

A Result Due to Gram-Schmidt Process

Theorem

The basis $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ constructed by the Gram-Schmidt process is an orthogonal basis of W . Moreover,

$$[\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_k] = [\mathbf{w}_1 \ \mathbf{w}_2 \ \dots \ \mathbf{w}_k]R,$$

where R is an upper triangular $k \times k$ matrix.

Structure of Matrix R

$$R = \begin{bmatrix} 1 & \frac{\langle w_1, v_2 \rangle}{\langle w_1, w_1 \rangle} & \frac{\langle w_1, v_3 \rangle}{\langle w_1, w_1 \rangle} & \dots & \frac{\langle w_1, v_k \rangle}{\langle w_1, w_1 \rangle} \\ 0 & 1 & \frac{\langle w_2, v_3 \rangle}{\langle w_2, w_2 \rangle} & \dots & \frac{\langle w_2, v_k \rangle}{\langle w_2, w_2 \rangle} \\ 0 & 0 & 1 & \dots & \frac{\langle w_3, v_k \rangle}{\langle w_3, w_3 \rangle} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

- Thus Gram–Schmidt leads naturally to a

QR factorization

QR Decomposition

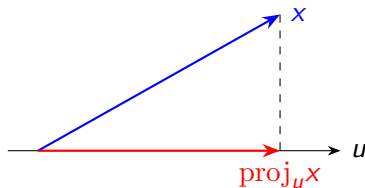
Theorem (QR Decomposition/Factorization)

Let A be an $m \times n$ matrix. If the columns of A are linearly independent, then A can be factorized as

$$A = QR$$

where Q is a $m \times n$ matrix whose columns form an orthonormal set, and R is an $n \times n$ upper triangular matrix whose diagonal entries are strictly positive.

Geometric Picture



- Subtracting projections enforces orthogonality.

Example

Example (GS Process)

Let $\mathbf{v}_1 = (1, 1, 0)$, $\mathbf{v}_2 = (1, 0, 1)$ in $\mathbb{R}^3$.

Solution. First, we have

$$\mathbf{u}_1 = (1, 1, 0).$$

Next,

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

Corollaries of GS Orthogonalization Process

- 1 Every finite-dimensional inner product space has an orthonormal basis.
- 2 Every orthonormal list of vectors in V can be extended to an orthonormal basis of V .

Summary

- (i) Inner products give geometry and hints us about the concept of angle.
- (ii) Norms measure length.
- (iii) Gram–Schmidt builds orthonormal bases.

Thank You

Questions?